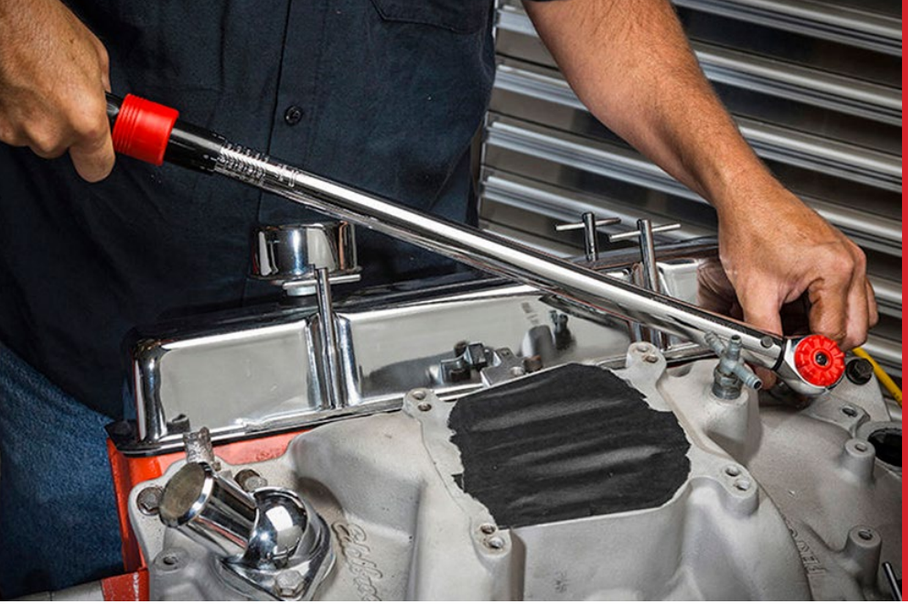


TORQUE **WRENCH**

FINAL PROJECT

Summary

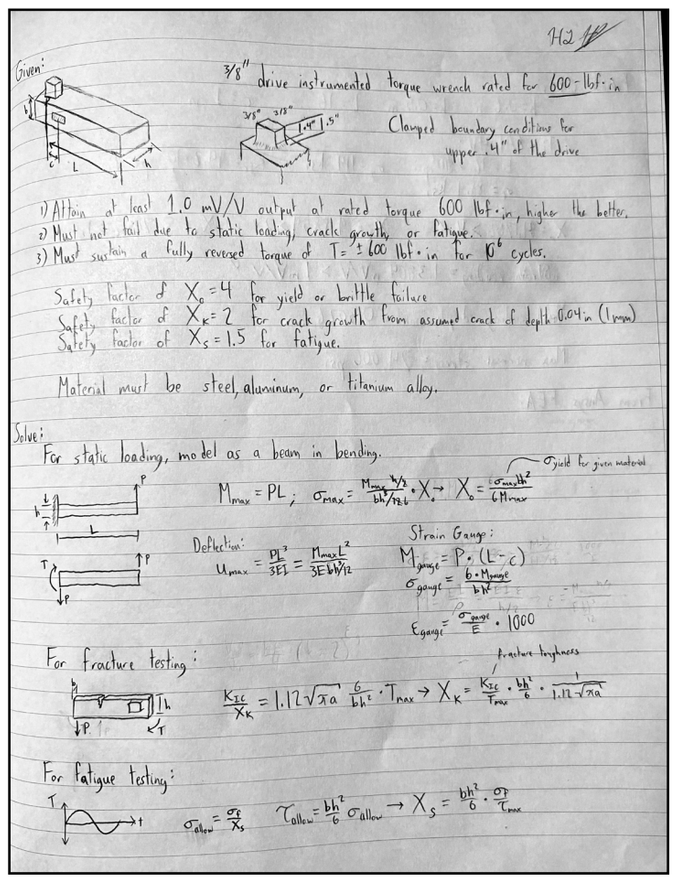
For the final project of MAE 3270: Mechanics of Materials, we were tasked with designing a simple torque wrench that could perform under specified parameters (static loading, fatigue, and fracture strength). After selecting the design and material that the torque wrench was made of, I needed to create a CAD and run an FEM over this model using ANSYS Static Structural. Comparing the analysis from ANSYS to my own hand calculations, I better understood the assumptions that had to be made to properly model the wrench.

Problem Statement

When beginning the preliminary design of the torque wrench, we were given the following problem statement:

"Your assignment is to design a non-ratcheting, 3/8 inch drive instrumented torque wrench rated for 600 in-lbf. Torque will be transduced using strain gauges bonded to the outer surfaces of the wrench at high strain locations. I am also asking you to perform a finite element analysis of your final design.

The torque will be transduced by strain gauges on the sides of the torque wrench. The design goal is to maximize the voltage output of the wrench (mV/V) at the rated torque. The design is required to attain at least 1.0 mV/V output at the rated torque of 600 in-lbf. Higher output will lead to more sensitivity and improved signal to noise ratio. The constrains are that the wrench must not fail due to static loading, crack growth or fatigue. Your design will include selecting an appropriate material and dimensions to meet or exceed the following requirements:



Given:

3/8" drive instrumented torque wrench rated for 600 lbf·in

Clamped boundary conditions for upper .14" of the drive

1) Attain at least 1.0 mV/V output at rated torque 600 lbf·in, higher the better.
2) Must not fail due to static loading, crack growth or fatigue.
3) Must sustain a fully reversed torque of $T = \pm 600$ lbf·in for 10^6 cycles.

Safety factor of $X_o = 4$ for yield or brittle failure
Safety factor of $X_K = 2$ for crack growth from assumed crack of depth 0.04 in (1 mm)
Safety factor of $X_s = 1.5$ for fatigue.

Material must be steel, aluminum, or titanium alloy.

Solve:

For static loading, model as a beam in bending.

$M_{max} = PL$; $\sigma_{max} = \frac{M_{max} \cdot h/2}{bh^3/12} \cdot X_o \rightarrow X_o = \frac{\sigma_{max} bh^2}{6 M_{max}}$ (σ_{max}: σ_{yield} for given material)

Deflection: $u_{max} = \frac{PL^3}{3EI} = \frac{M_{max} L^2}{3Ebh^3/12}$

Strain Gauge: $M_{gauge} = P \cdot (L - c)$, $\sigma_{gauge} = \frac{6 \cdot M_{gauge}}{bh^2}$, $\varepsilon_{gauge} = \frac{\sigma_{gauge}}{E} \cdot 1000$

For fracture testing:

$\frac{K_{Ic}}{X_K} = 1.12\sqrt{\pi a} \cdot \frac{6}{bh^2} \cdot T_{max} \rightarrow X_K = \frac{K_{Ic}}{T_{max}} \cdot \frac{bh^2}{6} \cdot \frac{1}{1.12\sqrt{\pi a}}$ (K_{Ic}: fracture toughness)

For fatigue testing:

$\sigma_{allow} = \frac{\sigma_f}{X_s}$ $\tau_{allow} = \frac{bh^2}{6} \sigma_{allow} \rightarrow X_s = \frac{bh^2}{6} \cdot \frac{\sigma_f}{\tau_{max}}$

- 1) Attain at least 1.0 mV/V output at the rated torque of 600 in-lbf.

- 2) Safety factor of X_0 = 4 for yield or brittle failure.

- 3) Safety factor of X_k = 2 for crack growth from an assumed crack of depth 0.04 inches (1 mm).

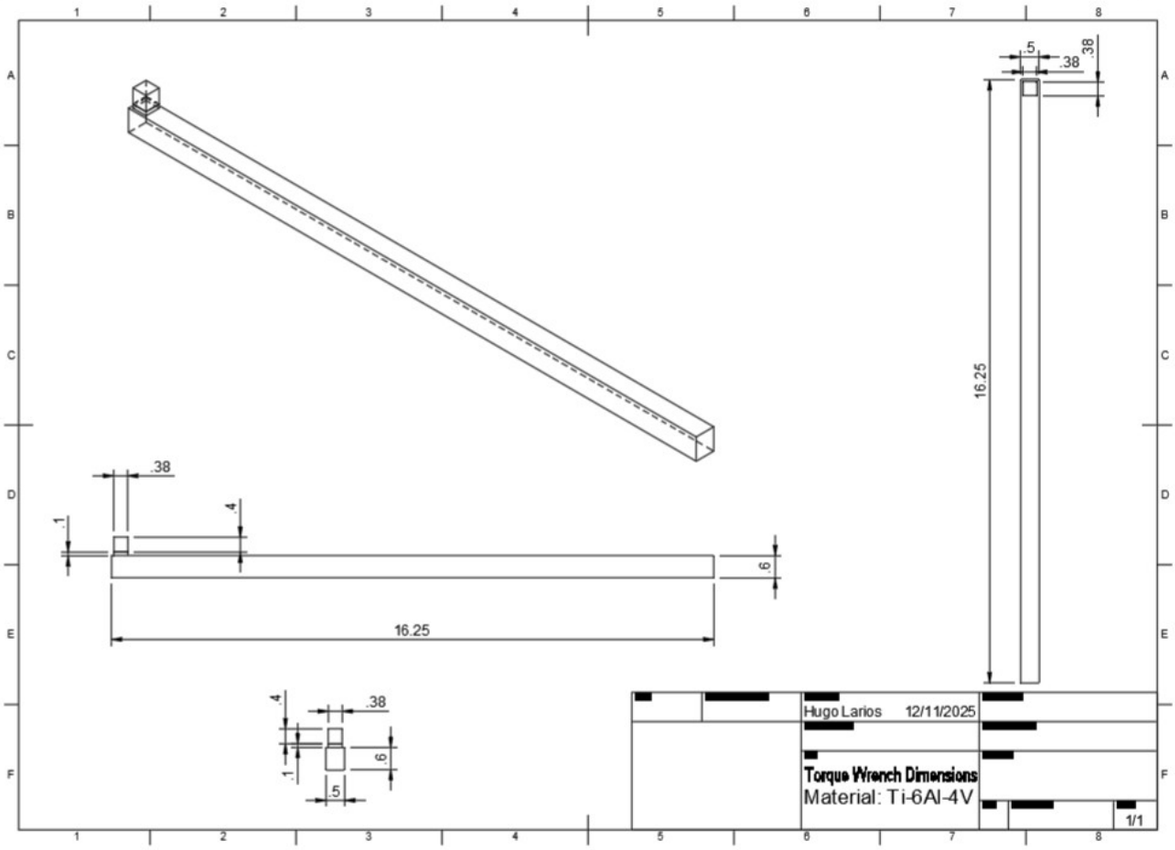
- 4) Fatigue stress safety factor of X_s = 1.5.

- 5) Material must be a steel, aluminum, or titanium alloy."

With these design considerations in mind, I first drew out the parameters and necessary equations within my notebook, before placing these equations into a MATLAB script. I first tested a steel alloy (M42 Steel), before moving onto a titanium alloy (Ti-6Al-4V).

MATLAB

```
b = 0.6;    h = 0.5;   L = 16;  % Dimensions of wrench (inches)
c = 1; % Location of strain gauge (inches)
X_o_in = 4;    X_k_in = 2;    X_s_in = 1.5; % Required Factors of Safety
M_max = 600; % Maximum allowable moment (lbf*in)
P = M_max / L; % Reaction force used for bending (lbf)
a = 0.04; % Crack length (inches)

% M42 Steel
% E = 32e+06; % Young's modulus (psi)
% stress_yield = 370e+03; % Yield strength (psi)
% KIC = 15e+03; % Fracture toughness (psi*sqrt(in))
% s_fatigue = 115e+03; % Fatigue strength for 10^6 cycles (psi)

% Ti-6Al-4V
E = 16.3e+06;   % Young's modulus (psi)
stress_yield = 114e+03;  % Yield strength (psi)
KIC = 94.2e+03; % Fracture toughness (psi*sqrt(in))
s_fatigue = 88.9e+03; % Fatigue strength for 10^6 cycles (psi)

% Static Loading
X_o = (stress_yield*b*(h^2)) / (M_max*6) % Static Factor of Safety

Max_normal_stress = (M_max) / (b*h^2 / 6) % Maximum normal stress

u_max = (M_max*L^2) / (3*E*b*h^3 / 12) % Max deflection (inches)

% Strain gauge calculations after bend
M_gauge = P*(L-c); % Moment felt at strain gauge
sigma_g = 6 * M_gauge / (b*h^2); % Stress felt at strain gauge

eps_g = sigma_g / E; % Calculation of microstrain
out = 1000 * eps_g % Conversion to mV/V

% Fracture Loading Factor of Safety
X_k = (KIC / M_max) * (b*h^2 / 6) * (1/(1.12*sqrt(pi*a)))

% Fatigue Loading Factor of Safety
X_s = (b*h^2 / 6) * (s_fatigue / M_max)
```



CAD Drawing

Above is the full drawing of my final CAD model. The critical dimensions for manufacturing and FEM modeling are listed.

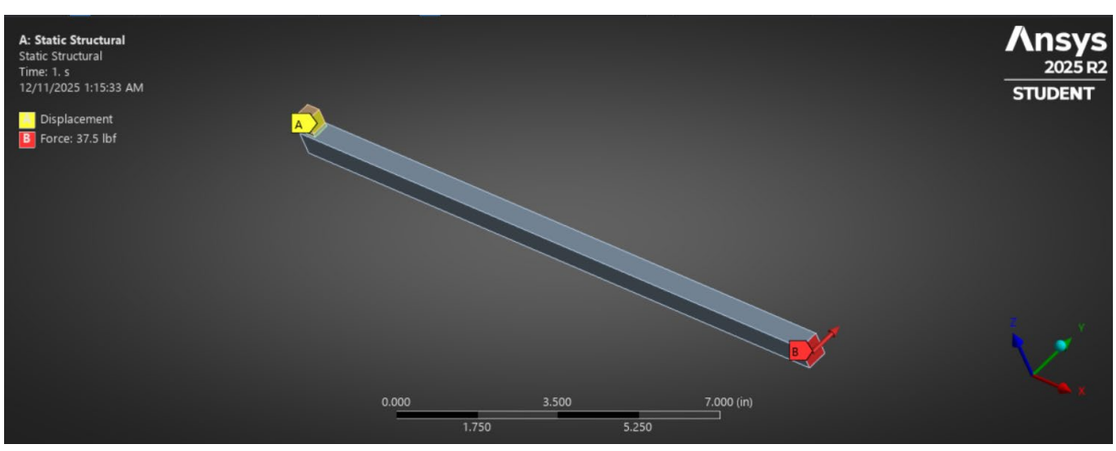
Material & Material Properties

The wrench is fabricated from Ti-6Al-4V, which best fits the material strain requirement.

The key properties include:
- Young's Modulus: 16.3*10^6 psi
- Poisson's Ratio: 0.332
- Yield Strength: 114 ksi
- Fracture Toughness: 94.2 ksi*sqrt(in)
- Fatigue Strength (10^6 Cycles): 88.9 ksi
- Density: 4.43 g/cm^3 (not structurally relevant but included for completeness)

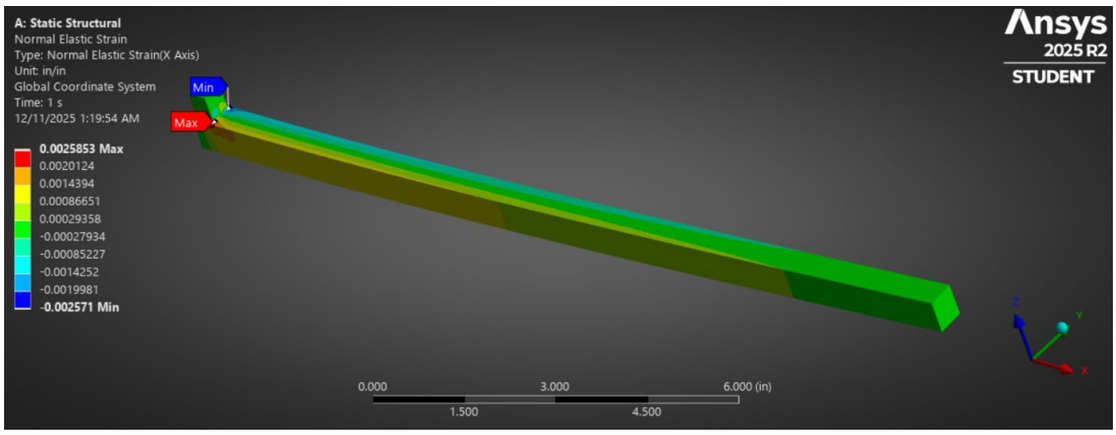
FEM Load and Boundary Condition Diagram

The FEM setup followed the course instructions and is illustrated below:
- The top 0.4 inches of the drive block was fully constrained (zero displacement, shown in yellow).
- A lateral force was applied on the wrench to create a net torque around the drive block. This force was F= 600/16 = 37.5 lbf.



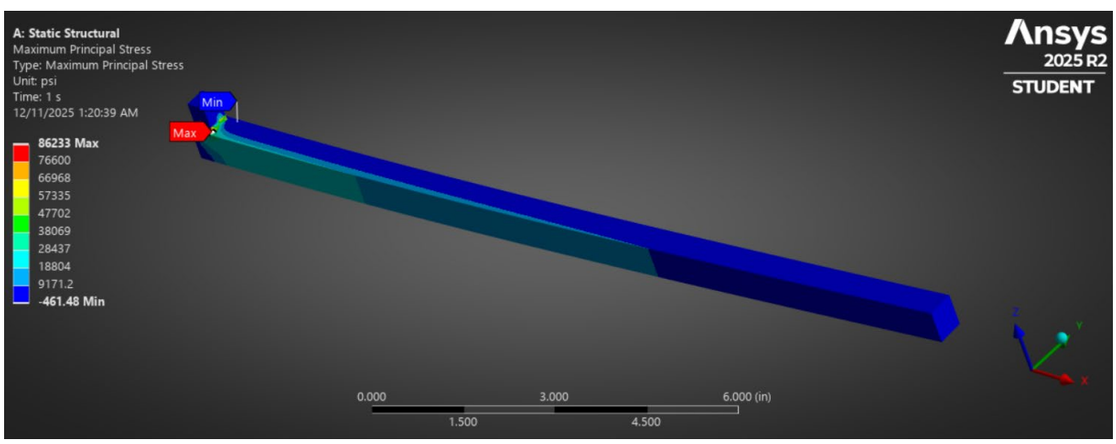
Normal Strain Contours (Gauge Direction)

Below is the strain distribution along the gauge axis. The peak gauge-direction strain from the probe was 2585.3 microstrain.



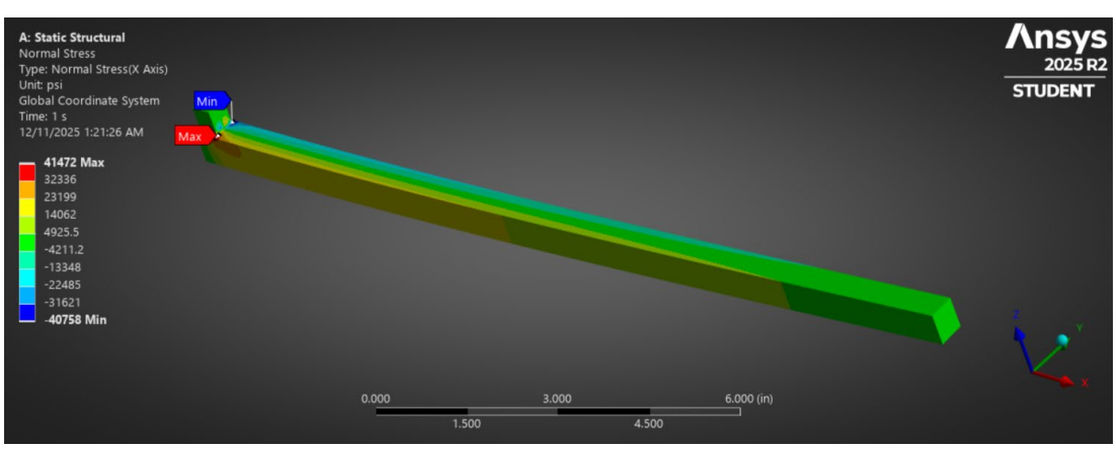
Maximum Principal Stress Contour

Maximum principle stress from the FEA was 86.223 ksi with a concentration of stress being found by the driver head.



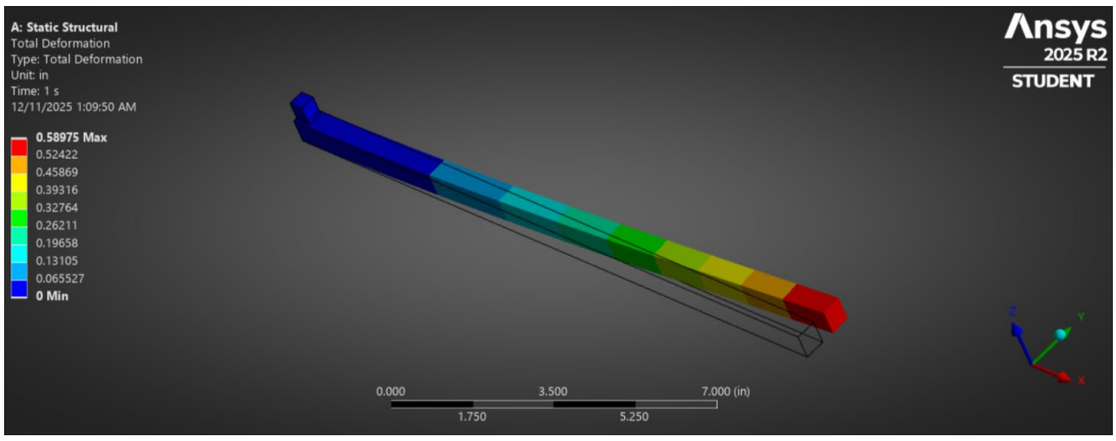
Maximum Normal Stress

There is a maximum normal stress of 41.472 ksi located at the driver head. This is to be expected, and the stress experienced here is significantly greater than the rest of the torque wrench.



Tip Deflection under a 600-lbf load

The maximum deflection that the torque wrench experiences is 0.58975 inch, which aligns well with the hand calculations.



Torque Wrench Sensitivity

Strain-gauge sensitivity is computed from Output(mV/V) = 10^3 * epsillon_gauge * GF. GF stands for the strain gauge factor, and in the case of using a half-bridge, this becomes GF = 1.0. This satisfies the geometry and yields a sensitivity of 1.3804 mV/V at 600 lbf-in in the Z-normal direction.

Strain Gauge Selection

A Micro-Measurements EA-06-031DE-120 linear foil gauge was identified. Its values were taken from the datasheet:
https://docs.micro-measurements.com/?id=2423
It had a length of 0.27 in and a width of 0.12 in.